

1 CONFIDENTIAL - DiabetesControl-X Development Program (DCX-505)

2 CLINICAL TRIAL PROTOCOL

3 Code Name: DiabetesControl-X (T2DM-505)

4 Study Title: A Phase 3, Multi-center, Randomized Trial for Type 2 Diabetes Mellitus

5 Sponsor: Global Pharma Solutions Inc.

6 Revision: 2.8

7 Date: March 10, 2026

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9 1. RATIONALE & OBJECTIVES

10 Type 2 Diabetes affects over 430 million people globally, with poor glycemic control in 50% of treated patients.

11 DiabetesControl-X offers superior HbA1c reduction and weight loss compared to standard GLP-1 agonists.

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13 Primary Objective: To demonstrate superiority of DiabetesControl-X vs. sitagliptin in HbA1c reduction.

14 Planned Target: Mean HbA1c reduction of 1.8% from baseline at Week 26.

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16 Secondary Objectives: Body weight reduction, cardiovascular safety, hypoglycemia rates.

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18 2. STUDY DESIGN

19 Double-blind, randomized, active-comparator trial across 60 global sites (North America, Europe, Asia-Pacific).

20 Enrollment: 1200 participants aged 18-75 with Type 2 Diabetes (HbA1c 7.5-11.0%).

21 Prior treatment: Inadequately controlled on metformin monotherapy or sulfonylureas.

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23 Study Duration: 26 weeks active treatment + 4 weeks safety follow-up.

24 Randomization: 1:1 ratio (DiabetesControl-X : sitagliptin).

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26 3. SAFETY MONITORING

27 Baseline assessments: HbA1c, fasting glucose, lipid panel, liver function tests, renal function, amylase/lipase.

28 Weekly home glucose monitoring during titration phase (weeks 0-4).

29 Bi-weekly remote monitoring via connected glucose meter for first 8 weeks.

30 Monthly in-clinic visits thereafter.

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32 Gastrointestinal Safety: Close monitoring for nausea, vomiting, pancreatitis.

33 Hypoglycemia Tracking: Participants instructed to report all hypoglycemic episodes via mobile app.

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Adverse Event Reporting: All SAEs reported within 24 hours.

Anticipated Effects: Nausea expected in 15-20% (typically mild, transient), weight loss in 100% of participants.

4. ENDPOINTS & STATISTICS

Primary Endpoint: Change in HbA1c from baseline to Week 26 (central laboratory).

Secondary Endpoints:

- Percent change in body weight from baseline
- Change in fasting plasma glucose
- Proportion achieving HbA1c <7% (<53 mmol/mol)
- Incidence of adverse gastrointestinal events
- Incidence of hypoglycemia (blood glucose <70 mg/dL)
- Change in cardiovascular markers (lipids, blood pressure, hsCRP)

Statistical Analysis: Intention-to-treat (ITT) population with last-observation-carried-forward (LOCF) imputation.

Sample size: 1200 provides 95% power to detect 0.4% HbA1c difference ($\alpha=0.05$, two-tailed).